MODULE 1: Introduction to Network Simulator and Sniffing Tools

1. What is a Computer Network?

A computer network is a collection of interconnected devices such as computers, servers, printers, and mobile devices that communicate and share resources through communication links.

Advantages:

- Resource sharing
- File sharing
- Internet access
- Communication

Example:

College computer lab connected through LAN.

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2. What is NS-3?

NS-3 (Network Simulator 3) is an open-source network simulation software used to study and analyze network behavior without creating a physical network.

Features:

- Supports wired and wireless networks
- Open source
- Supports C++ and Python
- Used in research and education

Uses:

- Performance testing
- Protocol analysis
- Network design

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3. Why do we use NS-3?

NS-3 helps simulate real network scenarios before actual deployment.

Benefits:

- Saves cost
- Saves time
- No hardware required
- Easy experimentation

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4. What is NetAnim?

NetAnim is a graphical animation tool used with NS-3.

Purpose:

It visually shows:

- Nodes
- Data transmission
- Packet movement

Benefit:

Makes simulation easy to understand.

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5. What is PyViz?
PyViz is a Python-based visualization tool for NS-3.
Functions:
- Shows network nodes
- Displays packet flow
- Visual debugging

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6. What is Wireshark?
Wireshark is a network protocol analyzer used to capture and inspect packets traveling through a network.
Uses:
- Network troubleshooting
- Security analysis
- Packet inspection

Features:
- GUI-based
- Real-time packet capture
- Supports many protocols

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7. What is Packet Sniffing?
Packet sniffing is the process of capturing and monitoring packets transmitted over a network.
Purpose:
- Analyze traffic
- Detect problems
- Security monitoring

Tools:
- Wireshark
- Tcpdump

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8. What is Tcpdump?
Tcpdump is a command-line network packet analyzer.
Advantages:
- Lightweight
- Fast
- Suitable for Linux servers

Example:
tcpdump -i eth0
Captures packets from interface eth0.

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9. Difference Between Wireshark and Tcpdump

Wireshark	**Tcpdump**
GUI Based	Command Line
User Friendly	Advanced Users
Detailed Visualization	Fast Capture
Easy Packet Analysis	Less Visual

Linux Commands Viva

10. What is Linux?

Linux is an open-source operating system based on Unix.

Features:

- Multi-user
- Multi-tasking
- Secure
- Free

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11. What does pwd command do?

pwd

Meaning:

Print Working Directory

Purpose:

Displays current directory path.

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12. What does ls command do?

ls

Lists files and folders in current directory.

Common options:

ls -l

Detailed listing.

ls -a

Shows hidden files.

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13. What is cd command?

cd foldername

Changes current directory.

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14. What is mkdir command?

mkdir test

Creates a new directory named test.

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15. What is rm command?

rm file.txt

Deletes a file.

rm -r folder
Deletes a directory recursively.

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16. What is ping command?
ping google.com
Tests connectivity between two devices.
Uses:
- Network troubleshooting
- Check host availability

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17. What is ifconfig?
Displays network interface information.
Shows:
- IP Address
- MAC Address
- Interface status

MODULE 2: Network Topology
18. What is Network Topology?
Network topology refers to the arrangement of devices and communication links in a network.
Types:
- Bus
- Star
- Ring
- Mesh
- Hybrid

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19. Explain Bus Topology.
All devices connect to a single communication cable called backbone.
Advantages:
- Low cost
- Easy installation

Disadvantages:
- Backbone failure affects entire network
- Difficult troubleshooting

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20. Explain Star Topology.
All devices connect to a central switch or hub.
Advantages:
- Easy maintenance
- Failure of one node doesn't affect others

Disadvantages:

- Switch failure affects network

Most commonly used topology.

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21. Explain Ring Topology.
Devices form a circular path.
Advantages:
- Organized communication

Disadvantages:
- One break can stop communication

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22. Explain Mesh Topology.
Every node is connected to every other node.
Advantages:
- High reliability
- Multiple paths available

Disadvantages:
- Expensive
- Complex

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23. Explain Hybrid Topology.
Combination of two or more topologies.
Example:
Star-Bus network.

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24. Which topology is most reliable?
Mesh topology because multiple communication paths exist.

MODULE 3: Basic Network Protocol Models
25. What is a Protocol?
A protocol is a set of rules that governs communication between devices.
Examples:
- TCP
- UDP
- HTTP
- FTP

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26. What is Client-Server Architecture?
A communication model where:
Client:
Requests services.
Server:
Provides services.
Example:

Web browser and web server.

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27. What is TCP?
TCP stands for Transmission Control Protocol.
Features:

- Reliable
- Connection-oriented
- Error detection
- Acknowledgement system

Uses:

- Web browsing
- Email
- File transfer

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28. What is UDP?
UDP stands for User Datagram Protocol.
Features:

- Fast
- Connectionless
- No acknowledgement

Uses:

- Video streaming
- Online gaming
- VoIP

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29. Difference Between TCP and UDP

TCP	UDP
Reliable	Unreliable
Connection Oriented	Connectionless
Slow	Fast
Uses ACK	No ACK

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30. What is FTP?
FTP stands for File Transfer Protocol.
Purpose:
Transfer files between client and server.
Port Number:
21

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31. What is HTTP?
HTTP stands for HyperText Transfer Protocol.
Purpose:

Transfer web pages.
Port Number:
80

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32. What is HTTPS?
HTTPS is secure HTTP using SSL/TLS encryption.
Port Number:
443
Advantage:
Secure communication.

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33. What is DHCP?
DHCP stands for Dynamic Host Configuration Protocol.
Function:
Automatically assigns:

- IP address
- Subnet mask
- Gateway
- DNS

Ports:
67 and 68

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34. What is DNS?
DNS stands for Domain Name System.
Function:
Converts domain names into IP addresses.
Example:
google.com → 142.x.x.x
Port:
53

---

35. What is SMTP?
SMTP stands for Simple Mail Transfer Protocol.
Purpose:
Sending emails.
Port:
25

MODULE 4: Network Traffic Analysis
36. What is Network Monitoring?
The process of observing network traffic and performance.
Goals:

- Detect issues

- Improve performance
- Ensure security

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37. What is Protocol Analysis?
Examining network packets to understand communication.
Tool:
Wireshark

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38. What is ARP?
ARP stands for Address Resolution Protocol.
Purpose:
Converts IP address into MAC address.
Example:
Finding MAC address of a device before sending data.

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39. What is MAC Address?
A unique hardware address assigned to a network interface.
Length:
48 bits
Example:
00:1A:2B:3C:4D:5E

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40. What is ICMP?
ICMP stands for Internet Control Message Protocol.
Uses:

- Error reporting
- Connectivity testing

Example:
Ping command

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41. What is Packet Filtering in Wireshark?
Filtering displays only selected packets.
Example:
http
Shows only HTTP packets.

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42. What is DNS Query?
A request sent to DNS server for domain resolution.

MODULE 5: Network Performance Evaluation
43. What is Network Performance?
Measurement of network efficiency and quality.
Parameters:

- Throughput
- Latency
- Packet loss
- Jitter

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44. What is Throughput?
Actual amount of data transferred successfully per second.
Unit:
Mbps or Gbps

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45. What is Bandwidth?
Maximum capacity of a network link.
Example:
100 Mbps network.

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46. Difference Between Throughput and Bandwidth
Bandwidth = Maximum capacity
Throughput = Actual achieved speed

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47. What is Latency?
Time taken for data to travel from source to destination.
Unit:
Milliseconds (ms)

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48. What is Packet Loss?
Packets that fail to reach destination.
Causes:

- Congestion
- Hardware failure
- Network errors

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49. What is Jitter?
Variation in packet arrival time.
Important for:

- Video calls
- Online gaming

MODULE 6: Real Time Network Problem Solving
50. What is Network Troubleshooting?
Process of identifying and resolving network issues.
Steps:

1. Identify problem
2. Analyze cause

3. Implement solution
4. Verify result

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51. How do you check connectivity?
Using:
ping

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52. How do you check routing information?
route
or
netstat -r

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53. How do you check network interfaces?
ifconfig
or
ip addr

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54. What is OMNeT++?
An open-source network simulation framework.

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55. What is QualNet?
Commercial network simulator for large-scale network analysis.

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56. What is NetSim?
Network simulation software used for academic and industrial purposes.

MOST IMPORTANT EXAMINER QUESTIONS
Explain OSI Model.

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Explain TCP/IP Model.

1. Network Access Layer
2. Internet Layer
3. Transport Layer
4. Application Layer

Difference Between IP Address and MAC Address.

IP Address	MAC Address
Logical Address	Physical Address
Can Change	Fixed
Layer 3	Layer 2

Which protocol does Ping use?
ICMP

Which protocol assigns IP addresses automatically?
DHCP

Which protocol converts domain names into IP addresses?
DNS

Which topology is most commonly used in companies?
Star Topology

Which topology provides highest fault tolerance?
Mesh Topology

ASSIGNMENT 1 (NS3 Installation)
Why do we install NS-3?
To simulate computer networks without creating physical networks.

Why is g++ installed?
Because NS-3 programs are written mainly in C++.

Why is Python installed?
NS-3 supports Python scripting and PyViz.

Why do we install Wireshark?
To analyze captured packets (.pcap files).

Why is tcpdump installed?
To capture and read packet traces from terminal.

What is a PCAP file?
Packet Capture file containing network traffic data.

Why is NetAnim installed?
To visualize packet movement graphically.

What does ./test.py do?
Tests whether NS-3 installation is successful.

What does
./waf --run hello-simulator
do?
Compiles and runs the hello-simulator program.

What is WAF?
WAF is NS-3's build system used to compile and run simulations.

ASSIGNMENT 2
POINT TO POINT TOPOLOGY
What is Point-to-Point topology?
A topology where two devices communicate using a dedicated link.

Why
nodes.Create(2);
?
Creates 2 nodes.

Examiner may ask:
What happens if you write
nodes.Create(5);
?
Five nodes will be created.

What is NodeContainer?
A container used to store network nodes.

Why
PointToPointHelper pointToPoint;
?
Creates a point-to-point communication channel.

Why
pointToPoint.SetDeviceAttribute("DataRate","5Mbps");
?
Sets transmission speed to 5 Mbps.

What is DataRate?
Maximum amount of data transferred per second.

What happens if DataRate becomes 10Mbps?
Transmission becomes faster.

Why
SetChannelAttribute("Delay","2ms");
?
Adds propagation delay of 2 milliseconds.

What is Propagation Delay?
Time taken by packet to travel from sender to receiver.

Why
InternetStackHelper stack;
?
Installs TCP/IP protocol stack.

Why
stack.Install(nodes);
?
Adds networking capability to nodes.

Why
Ipv4AddressHelper address;
?
Assigns IP addresses.

Why
address.SetBase("10.1.1.0","255.255.255.0");
?
Creates subnet 10.1.1.0/24.

Why use 255.255.255.0?
Subnet mask defining network and host portions.

What is
UdpEchoServerHelper
?
Creates a UDP Echo Server.

What is Echo Server?
Receives packet and sends same packet back.

Why Port 9?
Port 9 is traditionally used for Echo Service.

Why
serverApps.Start(Seconds(1.0));
?
Server starts at 1 second.

Why client starts at 2 seconds?
Server must start before client sends packets.

What happens if client starts first?
Packets may be lost because server isn't running.

Why
PacketSize = 1024
?
Each packet contains 1024 bytes.

Why
MaxPackets = 1
?

Client sends only one packet.

Why use UDP?
Simple communication with low overhead.

Why
MobilityHelper mobility;
?
Defines node movement or positions.

Why
ConstantPositionMobilityModel
?
Nodes remain stationary.

What is NetAnim XML file?
Stores animation data for visualization.

Why
AnimationInterface anim("first.xml");
?
Creates animation output file.

Why
EnablePcapAll("first");
?
Captures packets for Wireshark analysis.

What does Simulator::Run() do?
Starts simulation execution.

What does Simulator::Destroy() do?
Releases allocated resources.

BUS TOPOLOGY PROGRAM
How many nodes are created?
nNodes = 5
Five nodes.

Why is node 0 special?
Acts as central node connected to all others.

Why for loop starts from 1?

Node 0 is already used as central node.

What is
NodeContainer pair
?
Creates connection between two nodes.

Why
10.1.i.0
used?
Creates separate subnet for every connection.

Why use
PacketSinkHelper
?
Receives packets sent by clients.

What is Packet Sink?
Application that receives incoming packets.

What is
OnOffHelper
?
Traffic generator that alternates ON and OFF states.

Why
UdpSocketFactory
?
Uses UDP protocol.

What is
PopulateRoutingTables()
?
Automatically creates routing tables.

Why assign node positions?
To display topology correctly in NetAnim.

ASSIGNMENT 3
UDP CLIENT SERVER
What is UDP?
User Datagram Protocol.
Connectionless transport protocol.

Why is UDP called connectionless?
No connection establishment before communication.

Does UDP guarantee delivery?
No.

Why use UDP instead of TCP?
Lower overhead and faster communication.

Applications of UDP?

- DNS
- Video Streaming
- Gaming
- VoIP

Why
nodes.Create(2);
?
One client and one server.

Why use CSMA?
To simulate Ethernet LAN.

What is CSMA?
Carrier Sense Multiple Access.
Devices check channel before transmission.

Why
5Mbps
used?
Defines Ethernet speed.

Why
Delay = 2ms
?
Defines network latency.

Why
MTU = 1400
?
Maximum Transmission Unit size.

What is MTU?
Maximum packet size transferable without fragmentation.

Why enable PCAP?
Generate capture file for Wireshark.

Why
10.1.1.0
?
Network address.

What is
serverAddress
?
Stores destination IP of server.

Why port 4000?
Communication endpoint for client-server application.

What is a Port Number?
Logical communication endpoint.

Why
maxPacketSize = 1024
?
Each UDP packet contains 1024 bytes.

Why
maxPacketCount = 320
?
Client sends 320 packets.

What is
interPacketInterval = 0.05
?
50 ms delay between packets.

Why use Flow Monitor?
Measures network performance.

What statistics does Flow Monitor provide?
- Throughput
- Delay

- Packet Loss
- Sent Packets
- Received Packets

What is Throughput?
Actual successful data transfer rate.

How is throughput calculated?
Received bits ÷ transmission time.

Why serialize Flow Monitor results?
To save statistics in XML format.

VERY DANGEROUS EXAMINER QUESTIONS
Difference between Node and NodeContainer?
Node = Single device
NodeContainer = Collection of nodes

Difference between DataRate and Delay?
DataRate = Speed
Delay = Travel time

Difference between TCP and UDP?

TCP	UDP
Reliable	Unreliable
Connection Oriented	Connectionless
ACK Used	No ACK
Slow	Fast

Difference between NetAnim and Wireshark?
NetAnim → Visualization
Wireshark → Packet Analysis

Difference between PCAP and XML?
PCAP → Packet Capture
XML → NetAnim Visualization

Why do we use Simulator::Run() and Simulator::Destroy()?
Run executes simulation.
Destroy frees memory after completion.

Most Important Linux Commands for Viva
1. pwd
Command
pwd
Full Form
Print Working Directory
Use
Displays current directory path.

---

2. ls
Command
ls
Use
Lists files and folders.
Important Options
ls -l
Detailed information.
ls -a
Shows hidden files.
ls -la
Detailed + hidden files.

---

3. cd
Command
cd foldername
Use
Change directory.
Examples
cd Desktop
cd ..
Moves one directory back.

---

4. mkdir
Command
mkdir test
Use
Creates new directory.

---

5. rmdir
Command
rmdir test
Use
Deletes empty directory.

---

6. rm
Command
rm file.txt
Use
Deletes file.
Delete folder
rm -r foldername

---

7. cp
Command
cp file1.txt file2.txt
Use
Copies file.

---

8. mv
Command
mv old.txt new.txt
Use
Move or rename files.

---

9. cat
Command
cat file.txt
Use
Displays file contents.

---

10. touch
Command
touch file.txt
Use
Creates empty file.

---

11. clear
Command
clear
Use
Clears terminal screen.

Networking Commands
12. ifconfig
Command
ifconfig

Use
Displays network interface details.
Shows

- IP Address
- MAC Address
- Network Status

---

13. ip addr
Command
ip addr
Use
Modern replacement for ifconfig.

---

14. ping
Command
ping google.com
Use
Checks network connectivity.
Protocol Used
ICMP
Viva Question
Which protocol does ping use?
Answer: ICMP

---

15. traceroute
Command
traceroute google.com
Use
Shows path taken by packets.

---

16. netstat
Command
netstat
Use
Displays network connections.
Routing Table
netstat -r

---

17. route
Command
route
Use
Displays routing table.

---

18. arp
Command
arp -a
Use
Displays ARP table.
Viva Question
What is ARP?
Address Resolution Protocol.
Converts IP address to MAC address.

---

19. hostname
Command
hostname
Use
Displays system name.

---

20. nslookup
Command
nslookup google.com
Use
Finds IP address of domain.
Viva Question
Which protocol helps domain name resolution?
DNS

Wireshark Commands / Filters
Show HTTP packets
http

Show TCP packets
tcp

Show UDP packets
udp

Show DNS packets
dns

Show ARP packets
arp

Show ICMP packets

icmp

Show packets from specific IP
ip.addr == 10.1.1.1

Tcpdump Commands
Capture packets
tcpdump

Capture on specific interface
tcpdump -i eth0

Save capture
tcpdump -w capture.pcap

Read capture file
tcpdump -r capture.pcap

NS-3 Commands
Build NS-3
./build.py --enable-examples --enable-tests
Use
Compiles NS-3.

Test Installation
./test.py
Use
Checks whether NS-3 installed correctly.

Run Program
./waf --run hello-simulator
Use
Runs simulation.

Run Custom Program
./waf --run scratch/first
Use
Runs first.cc program.

NetAnim Commands
Open NetAnim
./NetAnim
Use

Launch NetAnim visualizer.

Commands Most Asked in Viva (Learn These First)

Command	Use
pwd	Current directory
ls	List files
cd	Change directory
mkdir	Create folder
rm	Delete file
cat	Display file
ping	Check connectivity
ifconfig	Show IP details
ip addr	Show network interfaces
arp -a	Show ARP table
netstat -r	Show routing table
nslookup	DNS lookup
tcpdump	Packet capture
./test.py	Test NS3
./waf --run hello-simulator	Run NS3 example
./NetAnim	Open NetAnim